WEEK 01

Values, types and execution

This week: learn, practise, build

01 · Python

Values, types and execution
Understand the idea and trace examples.

02 · Practice

Practise the same Python concept.
Predict, change, diagnose and explain.

03 · Project

Labelled planning calculation
Build, discuss and test with a peer.

One weekly component moves the final project forward.

Python foundations

Values, types and execution
Concept → trace a small example → explain its behavior

Names, values and statements

A value has a type

Integers support arithmetic; strings store text. Quoted text is a literal.

Assignment binds a name

The right side runs first. A later assignment changes what the name refers to.

Execution has an order

Run statements from top to bottom. Use print to observe a value at that point.

Understand the behavior before changing the example.

Python executes statements in order

1

Assign

count = 4
unit_size = 3
Names refer to values.

2

Evaluate

total = unit_size * count
Read values; multiply.

3

Use the result

print(total)
Display the stored value.

Assignment stores a value. Changing an input does not rerun an earlier expression.

Syntax to read and reuse

Python / syntax	Meaning	When to use it
name = value	Assign a value	Keep an input for later
3 * 4	Multiply numbers	Calculate a quantity
print("Total:", total)	Display a value	Label output and units
# explanation	Comment; not executed	Explain an assumption

Python Basics · sections 00, 01, 02, 03, 14 · use only when you need a lookup.

Names and text are different

```
01 count = 4
02 unit = 3
03 total = count * unit
04 print(count)
05 print("count")
06 print(type(count).__name__)
07 print(total)
```

1. A name retrieves its current value.
2. Quotes make text, not a variable lookup.
3. Expected: 4, count, int, 12.

Predict → execute → explain the value at each step.

Variables and string literals

```
containers = 4
print(containers)
print("containers")
print(type(containers).__name__)
```

1. Predict all three outputs before running
2. Assignment associates a name with a value
3. Change the value and predict again

Quotes create text; an unquoted name retrieves its assigned value.

Numbers and strings behave differently

```
01 print(3 * 4)
02 print("3" * 4)
03 assert 3 * 4 == 12
```

1. How does quoting the first value change the output?
2. Container counts use integer values
3. Keep expectations independent of output

Valid code can still express the wrong operation.

Assign, calculate and print

```
description = "Practice meal containers"
portions_per_container = 3
containers = 4
planned = portions_per_container * containers
print(description)
print("Containers:", containers)
print("Planned portions:", planned)
assert planned == 12
```

1. Assign input values first.
2. Use an expression to calculate a result.
3. print() displays the stored result.

Keep inputs, calculation and display as separate statements.

Execution order: when is a value recalculated?

```
01 portions_per_container = 3
02 containers = 4
03 planned = portions_per_container * containers
04 containers = 2
05 print("Before:", planned)
06 planned = portions_per_container * containers
07 print("After:", planned)
08 assert planned == 6
```

1. Predict the value printed before recalculation
2. Then predict the value printed after recalculation
3. Rerun the calculation after changing inputs

Changing an input does not rerun an earlier multiplication.

One language, two execution environments

Notebook: .ipynb

Narrative and code cells share a persistent kernel. Run cells and inspect their results.

Source file: .py

A new interpreter reads saved statements from top to bottom. Its output appears in a terminal.

A new script process does not inherit notebook variables.

Save and run a Python file

Context	Command or observation
Working directory	learning_materials/
Terminal	python programs/first_pickup.py
Choose Python	Use your course interpreter; see programs/README.md
Default calculation	3 portions/container × 4 containers
Expected meaning	12 planned portions, not delivered impact

Replace python with your course interpreter; save before rerunning.

Hands-on practice

Week_01.ipynb · Session 2
Predict → run → change → explain

Your turn: practise the same Python skills

Predict values

Read count and "count". Predict the type and product. Run.

Change and observe

Change count after total. Explain the old result; recalculate.

Repair a type bug

"4" * 3 repeats text. Make the result numeric; explain why.

Use the matching exercises in this weekly notebook. Explain what changed and why.

Check your work before asking for help

Predict first

Write what you expect before executing.

Change one thing

Compare values, types, paths or returned results.

Explain the cause

Use the relevant Python rule. Retest your repair.

Explain names versus values, number/string operations and execution order.

Project studio

Labelled planning calculation
Short design briefing → most time for your work

The people’s task this week

1

Someone needs help

The coordinator asks: how much are we planning?

2

This week, you will

Use the declared counts to calculate and label planned portions.

3

A useful result

The coordinator can read the estimate and ask about unclear units.

PICTURE IT: 4 containers × 3 = 12 planned portions.

Use the supplied starting files. Focus on one useful result and check it.

Trace a declaration into a planning note

1

Declared input

3 planned portions in each specified container

2

Proposed count

4 containers in this practice plan

3

Calculation

3 multiplied by 4 gives 12 planned portions

4

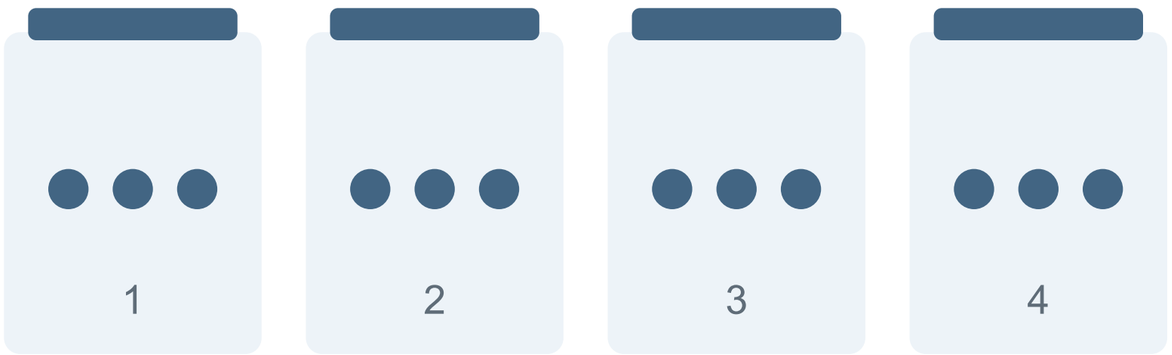
Labelled output

Show the units and retain planning status

Portions per container multiplied by containers gives planned portions.

See the values behind the calculation

4 containers



3 declared portions per container

3 × 4 = 12

Planned portions

A calculation from declared inputs.

Read the diagram, then trace the same steps in code.

Your implementation TODO

Supplied starting point

A notebook starter provides variable names and print calls.

You implement / do

Set description, unit size 2 and containers. Calculate and display planned_portions.

Check the evidence

Run containers 1, 3 and 0: expect 2, 6 and 0. Explain why this is a plan, not recorded delivery.

Explain the input units, the calculation and one coordinator question.

Show the component working

Step	What to show
1 · Locate	Identify the input, this component and its next user.
2 · Implement	Print description, input counts and planned portions.
3 · Check	2 portions/container × 1, 3, 0 containers → 2, 6, 0.
4 · Explain	Show your code, its actual result and why it behaves that way.

Explain the input units, the calculation and one coordinator question.

Further reading in English

1. ## Python tutorial: numbers and strings

   https://docs.python.org/3/tutorial/introduction.html

2. ## Python tutorial: input and output

   https://docs.python.org/3/tutorial/inputoutput.html

Read one relevant section, then bring an example and a question.